

PAPER_941: Linewidth Jet Modulation Engine

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** linewidth_jet_modulation.py (LinewidthJetModulationSweep) **Calculator:** LinewidthJetModulationSweepCalc (CP4 #525) **CVW:** v2.0.0 compliant

Abstract

We present a systematic linewidth-to-jet modulation mapping engine that sweeps Γ from 0.01 to 1.0 THz and computes the jet modulation factor M_{jet} , quality factor Q , and operational regime (narrow / optimal / broad) at each point. The engine provides a universal lookup for any astrophysical jet system, decoupling the linewidth physics from specific source parameters.

1. Core Equations

$$M_{\text{jet}}(\Gamma) = 1 + A_{\text{jet}} \exp\left(-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right) \cdot S_{26} \cdot \left(\frac{2F_{U\text{Bi}}}{F_U} - 1\right)$$
$$Q = \frac{\omega_{\text{SCm}}}{2\Gamma}$$

where $S_{26} = \sum_{k=1}^{26} e^{-[\text{SSq}] \cdot k/26}$ and $[\text{SSq}] = 0.57$.

2. Regime Classification

Regime	Γ Range	Characteristics
Narrow	$\Gamma \leq 0.07$ THz	High Q , tight collimation
Optimal	$0.07 < \Gamma \leq 0.15$ THz	Peak modulation
Broad	$\Gamma > 0.15$ THz	Low Q , wide opening angle

3. Reference Systems

The ReferenceSystemMatcher class compares computed (M_{jet}, Q) pairs against five calibrated systems: M87 ($A_{\text{jet}} = 1.20$), Sgr A* (0.80), Centaurus A (0.95), TXS 0506+056 (1.20), and 3C 273 (1.05).

4. Source Data

- **File:** linewidth_jet_modulation.py
 - **Session:** 213
 - **CP4 Class:** LinewidthJetModulationSweepCalc (#525)
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References

1. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)
2. Blandford, R.D. & Konigl, A. (1979) - ApJ, 232, 34